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Smart Pill Box using IoT and Cloud

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ABSTRACT

This paper focuses on to provide smart service and aid to elderly patients or patients who generally have a problem with adhereing to medications for a prolonged period of time. In this proposed work, there are three systems. Firstly, Reminder System which helps the patient to take the required medications on time. The patient gets a text SMS on a mobile device 15 minutes prior to medication time. If the box is opened, the caretaker gets a text message on this phone. This helps the patient to take medications requlary and conveniently. Secondly, Medication Monitoring System which provides the caretaker and the doctor with data and logs about the patient's medications. The data is retrieved from sensors in the Pill Box and stored on the cloud. The caretaker or the doctor can retrieve this data to see if the patient is actually taking his medications on time. And lastly, Medication Inventory System which helps the patient to keep a track of his supply of medicines. The patient gets an SMS on phone when the Smart Pill Box has a low supply of medications and may run out of medicines soon. This helps the patients to purchase medications before they run out, so that they do not miss any dosage.The work was implemented using IoT and cloud technologies. The results have proved that the system's performance is reliable.

KEY WORDS: MEDICATION MONITORING SYSTEM, SENSORS, CLOUD, IOT.

INTRODUCTION

Most of the technology we use today is based on convenience, for example, vehicles for the easy mode of transportation, bullet train for quicker and faster traveling, mobiles for communication, gathering information and these devices still keep on advancing a step further for much better user experiences. Convenience is another phrase for 'time saver', and into the day's world which is developing and moving ahead at a very fast pace, each and every second has value. Smaller conveniences at home would be desirable because they will allow us to save time and monitor the patient both at the same time.

There are several medicine dispensing systems used in hospitals but the idea of getting a medicine dispensing box at home will take the comfort the patients to a whole new level. Smart Pill Box assures the patient to take their pills on time as well as gives them the authority to control their inventory. In case a patient forgets to take his/her medication the cloud based connectivity can help the patients to retrieve their information and statistics in a proper form. The hospital which is associated with

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the patient can also send immediate supplies as they would be aware that the inventory is getting depleted. Moreover, the doctors would know if the patient is being administered medicines on time and this would in turn help the patients to be healthy.

The health sector is expanding every day and now people are busy with their scheduled life. At this stage everything that reduces concern about something can be regarded as a boon and the Smart Pill Box plays along exactly the same lines. Patients who have already been discharged from the hospitals still need the same amount of care as they needed when they were admitted. The Smart Pill Box can reduce the concern of the immediate family members to great extent.

MATERIAL AND METHOD

Literature review: (Shinde et al,. 2017; Jayakumar et al,. 2015) in his work proposed a smart pillbox with remind and consumption function, which is used to give alert the user to take pills at a particular time and the pills required to take at that time comes out to the user to avoid confusion among medicines.Smart pill box can reduce elderly family member's responsibility towards giving the correct and timely consumption of medicines. This system gets the feedback about pills from the user and send purchase order to medical shop.

(Naga et al,. 2018; Jayakumar et al,. 2015)in her paper proposed a smart med box on which there will be a display which notifies the people about the medicine. Along with this we can alert them with an alarm and light indications. So that even if the person is sleeping or busy with some work the alarm helps in alerting him to confirm that the person has taken that medicine or not. There is a button at the opening end of the pill box, so when the person tries to open the box the button is pressed and the alarm will be off only if the buzzer is pressed.The design proposed by (Aakash et al 2015; Jayakumar et al,. 2017help the patients to take the required medicine in the right proportion at the right time. The basic ideology is integrating the principle of Alarm clock with Light based slot sensing on a normal pill box. An alternate to the light based sensing method using capacitive fields is also employed.

To make it more state-of-the- art, it is inbuilt with a GSM module for alerting the patient and also the chemist at the needed instant.) (Sanjay et al,. 2017;Jayakumar et al,. 2018)paper proposes a Smart medicine box for those users who regularly take medicines and the prescription of their medicine is very long as it is hard to remember to patients and also for their care giver. These problems persist in hospitals & people around us who have such kind of diseases and thus based on these two problems a smart medicine box is proposed which solve these problems by Setting up time table of prescribed medicines through push buttons as given in prescription.

(Wu et ala,. 2015; Jayakumaret al,. 2019) proposed pill box, where the camera is placed in inner side of the box to detect the matrix barcode and the medicine bag. A hardware module above the box was used to provide pill reminding and alarm functions. After visiting a doctor and returning home, a patient need only scans the matrix barcode using the camera of the pillbox, and all medicine related information will be loaded into the pill box. After the matrix barcode is scanned, the patient places the medicine bags in the pill box without dispensing the medicine in to the cell.

(Huang et al,. 2014; Lokeshkumar et al,.2019) proposed the smart pill box in 2012. The purpose of this system was to develop a medication device that increased medication compliance, monitored medication taking behavior,and communicated with pharmacists.The device consists of 28 chambers that are placed in seven columns made up of four rows. Each column represents each day of a week. The 4 rows represent four distinct dosage times in a day. The LED light located behind each chamber provides the light source for the patients. The light also used as the indication for the ambient light sensor which is used to detect the light when the pills are removed from the chamber.

(Abbey et al,. 2012; Jafar et al,. 2019) proposed a system where the infrared sensors are fixed at the entering where the patients take the medicine package.The detection of medicine taken will be delivered to the back-end control system and record to SQL server via the wireless serial port. The time to take the medicine package away will be recorded. The motor controls the spring which is used to put the medicine packages shown as figure. A base board is installed under the spring. As the motor rotates, the packages will be pushed forward. A package drops to the entrance when it moves out of the range of the board.

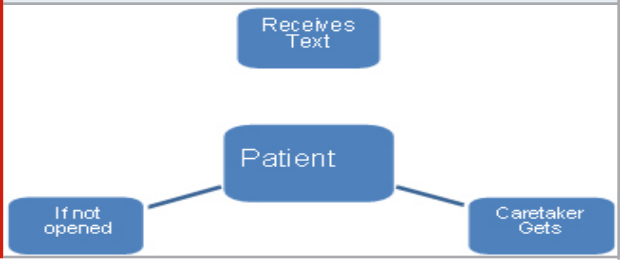
Proposed System

The smart bill box consists of three parts:

- Reminder System
- Patient Monitoring System
- Inventory System

Reminder System: The reminder system in the Smart Pill Box uses a messging API to send text messages to the patient and the caretaker. The patient gets a message on the mobile 15 minutes prior to the time when the medicine is supposed to be taken. After getting the SMS, the patient can open the Smart Pill Box and take the medications.

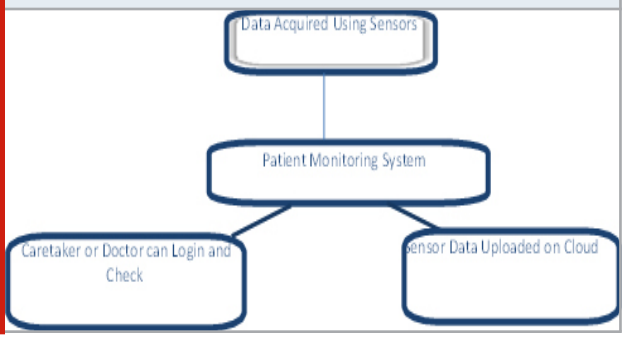
Figure 1: Flow Chart of Reminder System



When the patient takes the medication and the box is opened, an SMS alert is sent to the caretaker stating that the patient has taken the medication. So, if the caretaker has not received any notification, he/she can call the patient and remind him to take the medications.

Patient Monitoring System: This section allows the caretaker and the doctor to keep track of if the patient is really taking the medication. IR module is used to get data if the Smart Pill Box was opened.

Figure 2.2. Flow Chart of Patient Monitoring System



Then this data is Uploaded on the Cloud. The cloud service used here is IoT Makers. It fetches data from the device in real-time and updates on the cloud.
IR Sensor Module

The IR sensor module is attached to the Smart Pill Box. This sensor provides data if the box was opened or not along with a time-stamp. This data, along with the time-stamp is uploaded on the cloud. The IR sensor data is also used to send an SMS alert to the caretaker if the box was opened.

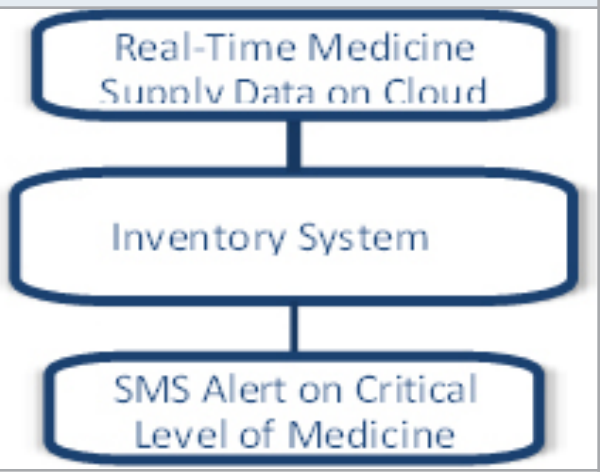
ESP8266 Wi-Fi Module: This is the mode of internet connectivity to achieve functions related to the cloud. An active Wi-Fi internet connection is required to send data to the Cloud and update. The ESP8266 connects to a Wi-Fi network and provides internet connectivity.

IoT Makers Cloud: This is an IoT based platform for uploading real-time sensor data to the cloud. The data is uploaded along with a time-stamp. The data includes if the box was opened. And the time at which the box was opened. This allowes the caretaker and the doctor to keep track of the medications of the patient.

Inventory System: This system is used to keep a track of the medicine supply, so that medicine dosages are not missed when the Smart Pill Box runs out of medicine. This uses an Ultrasonic Sensor to detect the level of medicine in the box. This data is uploaded live on the cloud. The caretaker and the doctor can check it on the Web Page in real- time to get an idea about when the box may run out of pills.

This system also sends an SMS alert on the Mobile Phone of the patient when the box runs out of medicine. When the sensor senses a critically low level of medicine present in the box, it alerts the GSM module, and an SMS alert is sent to the patient. This allows the patient to restock the supply of medicines before they run out. And this in-turn allows the patient to never skip medicines due to unavailability of medicine supply.

Figure 2.3: Flow Chart of Inventory System



RESULTS AND DISCUSSION

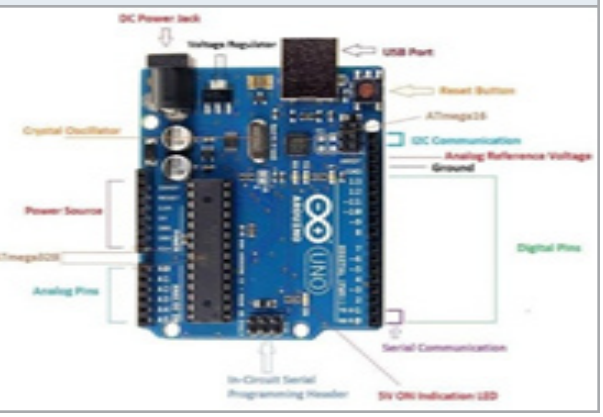
Implementation

The implementationinvolves various sensors and devices namely:

- Arduino UNO
- IR Sensor Module
- Ultrasonic Sensor
- Wi-Fi ESP8266 Module
- Arduino SIM800C GSM Module
- Medicine Board

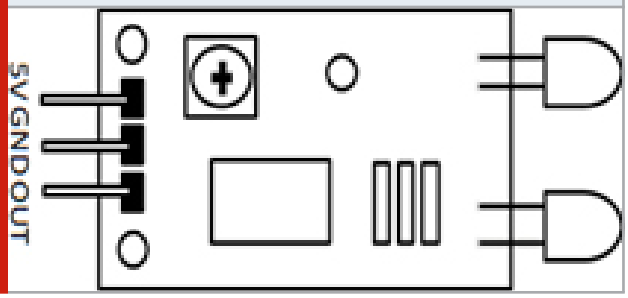
Arduino Uno: It is an open-source based micro-controller developed by Arduino.cc i.e., it can be configured by the user according to the application, based on Atmega328. It is introduced to minimize the effort on the input side and maximize the output. Microcontrollers are an important part of electronic and automation industry, which are widely used in the embedded system to make the user's task much easier and time-saving.

Figure: Arduino UNO



IR Sensor Module: There are three basic components in an IR sensor module. They are: the IR transmitter, the IR receiver and the LM385 OpAmp amplifier. The basic function of using an IR sensor module is to sense obstacles using the transmitter and the receiver.

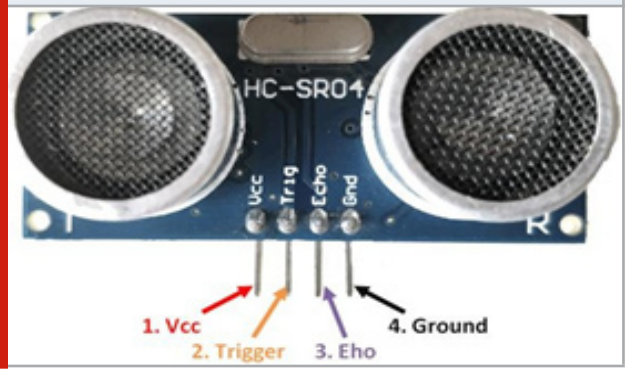
Figure 3.5. IR Sensor Module (Image and PIN Configurtion)



The transmitter in the IR sensor module transmits light in the range of infrared frequency. This light bounces back when it hits an obstacle. This light is then received on the receiver. This allows the module to detect any obstacles that may be present. After an obstacle is encountered, the amplifier can change the voltage in the I/O pins as low or high. When no obstacle is present, low signal is sent to I/O. When an obstacle is present, high signal is sent to the I/O.

Ultrasonic Sensor: Ultrasonic sensors mainly are used for object detection. In this model, we have used it for two different purposes, the first one for obstacle avoidance and secondly for Urine level detection. The used ultrasonic module has a range of 2cm to 450cm, has a resolution of 0.3cm, the effectual angle of <15 degree and quiescent current in it is <2mA.

Figure 3.6: Ultrasonic Module



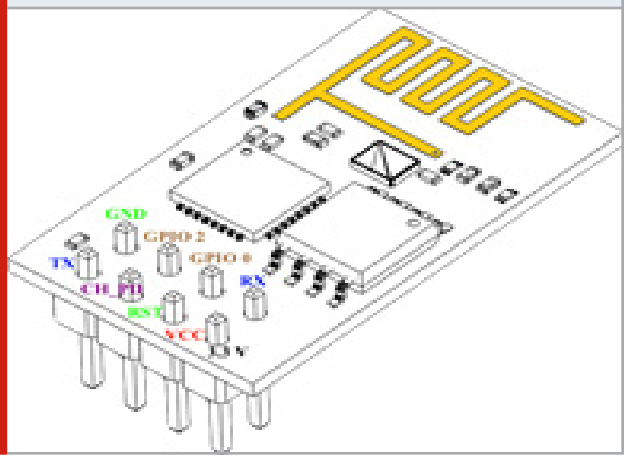
It has an operating voltage level of +5V. It consists of a transmitter, a receiver and control circuit. The transmitter transmits eight 40kHz ultrasonic waves by triggering IO for at least 10us high-level signal. If there is an object in the path of waves then it will bounce back to the module and is received by the receiver. Traveled distance by waves can be calculated by considering the traveling time of waves and the known speed of sound.

Wi-Fi ESP8266 Module: The Wi-Fi ESP8266 module is used to connect the Smart Pill Box to the internet. This connects to the internet through a Wi-FI network.

Table 3.4. Ultrasonic Sensor PIN Configuration

Pin Number	Pin Name	Description
1	Vcc	Vcc is used to provide 5V power to the sensor.
2	Trigger	Transmits eight 40kHz ultrasonic waves by triggering IO for at least 10us high-level signal
3	Echo	Echo pin is an Output pin.
4	Ground	This pin is used to connect to Ground.

Figure 3.7: ESP 8266 PIN Configuration



- VCC, Voltage (+3.3 V; can handle up to 3.6 V)
- GND, Ground (0 V)
- RX, Receive data bit X
- TX, Transmit data bit X
- CH_PD, Chip power-down
- RST, Reset
- GPIO 0
- GPIO 2

The ESP8266 is paired with the Arduino to help the microcontroller connect to the internet. This internet connectivity is essential to achieve cloud functionality. SIM800C GSM Module

Figure 3.8: GSM Module SIM800C



The SIM800C GSM module is paired with the Arduino to implement GSM connectivity. The GSM module allows the Smart Pill Box to send SMS alerts to mobile devices for better connectivity. This module is used to send alerts to the caretaker when the box is opened. This is also used to send urgent alert when the level of medicine in the box is critically low.

Figure 5.1: IR Sensor Module



Figure 5.2: Ultrasonic Sensor

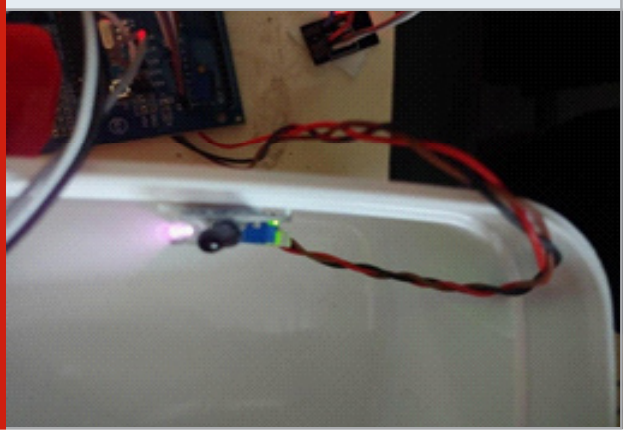
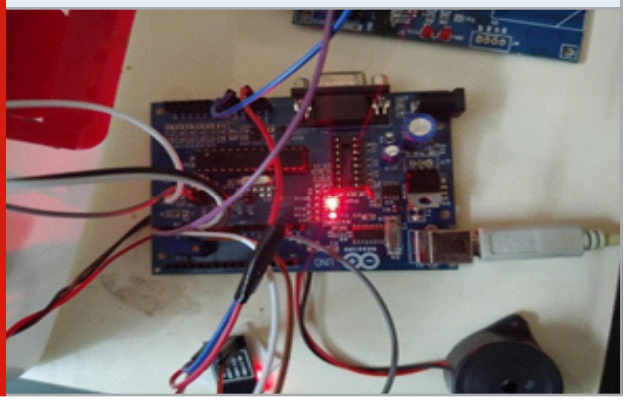
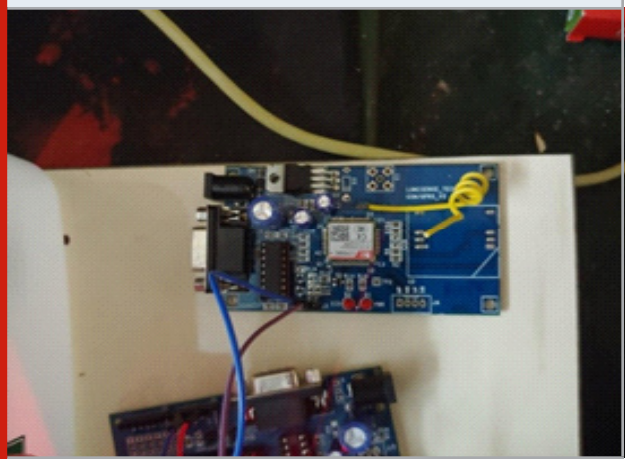


Figure 5.3: Arduino Uno in I/O Mode



Reminder System: The reminder system in the Smart Pill Box uses a messging API to send text messages to the patient and the caretaker. The patient gets a message on the mobile 15 minutes prior to the time when the medicine is supposed to be taken. After getting the SMS, the patient can open the Smart Pill Box and take the medications. (fig 5.4) When the patient takes the medication and the box is opened, an SMS alert is sent to the caretaker stating that the patient has taken the medication. So, if the caretaker has not received any notification, he/she can call the patient and remind him to take the medications.

Figure 5.4: GSM SIM800C Module



Patient Monitoring System: This section allows the caretaker and the doctor to keep track of if the patient is really taking the medication. IR module (fig 5.1) is used to get data if the Smart Pill Box was opened. Then this data is Uploaded on the Cloud. The cloud service used here is IoT Makers. It fetches data from the device in real-time and updates on the cloud.The IR sensor module is attached to the Smart Pill Box. This sensor provides data if the box was opened or not along with a time-stamp. This data, along with the time-stamp is uploaded on the cloud. The IR sensor data is also used to send an SMS alert to the caretaker if the box was opened.The ESP8266 is used for internet connectivity to achieve functions related to the cloud. An active Wi-Fi internet connection is required to send data to the Cloud and update. The ESP8266 connects to a Wi-Fi network and provides internet connectivity. IoT Makers is used to store real-time data on the cloud. The data is uploaded along with a time-stamp. The data includes if the box was opened. And the time at which the box was opened. This allowes the caretaker and the doctor to keep track of the medications of the patient

Inventory System: This system is used to keep a track of the medicine supply, so that medicine dosages are not missed when the Smart Pill Box runs out of medicine. This uses an Ultrasonic Sensor(fig 5.2) to detect the level of medicine in the box. This data is uploaded live on the cloud. The caretaker and the doctor can check it on the Web Page in real-time to get an idea about when the box may run out of pills.It also sends an SMS alert on the Mobile Phone of the patient when the box runs out of medicine. When the sensor senses a critically low level of medicine present in the box, it alerts the GSM module(fig 5.3), and an SMS alert is sent to the patient. This allows the patient to restock the supply of medicines before they run out. And this in-turn allows the patient to never

skip medicines due to unavailability of medicine supply. After development, remodeling and testing the product in different situations and conditions, detailed system wise results are provided in this section. Comments on obtained data, observed behavior, working accuracy, and reliability are also incorporated.

Reminder System: Below figures show the data obtained through an SMS alert seen by the patient and the caretaker

Figure 5.5: When the patient opens the box and takes the medicine

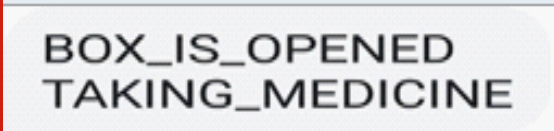
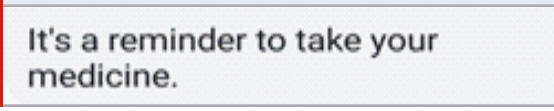


Fig 5.6: Reminder received on the Patient End



Figure

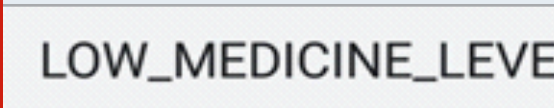


Fig 5.5 shows the text message received by the caretaker when the patient opens the box and takes the medicine. The accuracy of the IR Sensor used is 70-80%. Fig 5.6 shows an SMS alert received on Patient End to take the medications. Fig 5.7. Alert received by the patient when the Smart Pill Box has less medicines

Fig 5.7 shows the alert received by the patient when the level of medicines in the Smart Pill Box is critically low and may run out soon. The accuracy of the ultrasonic sensor is around 60-70%.

5.5Patient Monitoring System: This consists of the real-time cloud environment where the data gets uploaded and can be monitored by the caretaker and the doctor.

Figure 5.7: The Cloud Environment where the Data gets uploaded in real-time

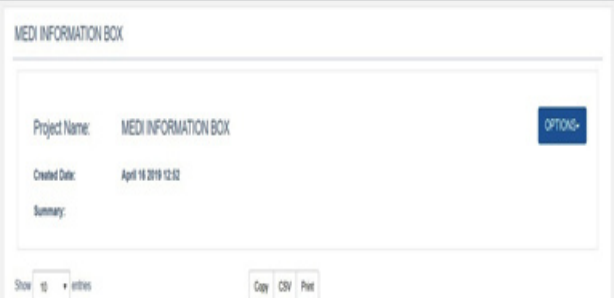
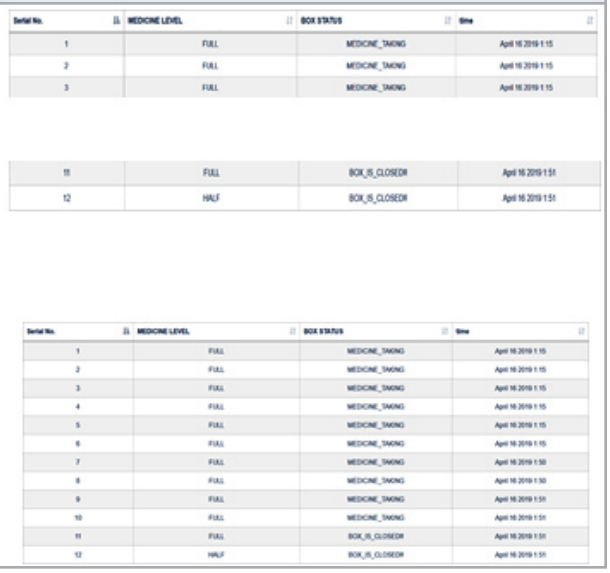


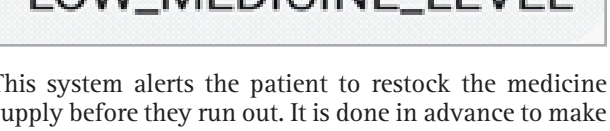
Fig 5.8: Medicine Level, Status of Box and Time-Stamp Inventory System



Serial No.	MEDICINE LEVEL	BOX STATUS	time
1	FULL	MEDICINE_TAKING	April 16 2019 1:15
2	FULL	MEDICINE_TAKING	April 16 2019 1:15
3	FULL	MEDICINE_TAKING	April 16 2019 1:15
11	FULL	BOX_IS_CLOSED!!	April 16 2019 1:51
12	HALF	BOX_IS_CLOSED!!	April 16 2019 1:51

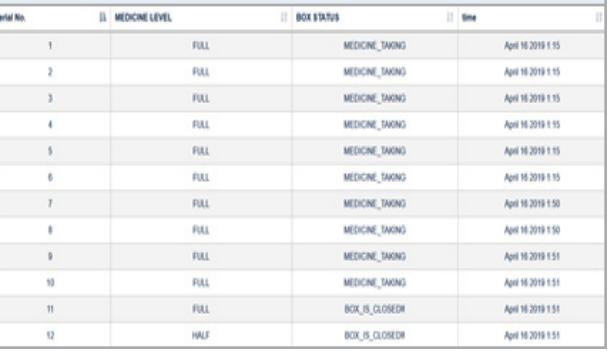
Serial No.	MEDICINE LEVEL	BOX STATUS	time
1	FULL	MEDICINE_TAKING	April 16 2019 1:15
2	FULL	MEDICINE_TAKING	April 16 2019 1:15
3	FULL	MEDICINE_TAKING	April 16 2019 1:15
4	FULL	MEDICINE_TAKING	April 16 2019 1:15
5	FULL	MEDICINE_TAKING	April 16 2019 1:15
6	FULL	MEDICINE_TAKING	April 16 2019 1:15
7	FULL	MEDICINE_TAKING	April 16 2019 1:50
8	FULL	MEDICINE_TAKING	April 16 2019 1:50
9	FULL	MEDICINE_TAKING	April 16 2019 1:51
10	FULL	MEDICINE_TAKING	April 16 2019 1:51
11	FULL	BOX_IS_CLOSED!!	April 16 2019 1:51
12	HALF	BOX_IS_CLOSED!!	April 16 2019 1:51

Figure 5.9: Alert to restock the medicine supply



This system alerts the patient to restock the medicine supply before they run out. It is done in advance to make sure that there is enough time to check the availability of medicines in the market and purchase them.

Figure 7: Snapshot of forecasting prototype



Serial No.	MEDICINE LEVEL	BOX STATUS	time
1	FULL	MEDICINE_TAKING	April 16 2019 1:15
2	FULL	MEDICINE_TAKING	April 16 2019 1:15
3	FULL	MEDICINE_TAKING	April 16 2019 1:15
4	FULL	MEDICINE_TAKING	April 16 2019 1:15
5	FULL	MEDICINE_TAKING	April 16 2019 1:15
6	FULL	MEDICINE_TAKING	April 16 2019 1:15
7	FULL	MEDICINE_TAKING	April 16 2019 1:50
8	FULL	MEDICINE_TAKING	April 16 2019 1:50
9	FULL	MEDICINE_TAKING	April 16 2019 1:51
10	FULL	MEDICINE_TAKING	April 16 2019 1:51
11	FULL	BOX_IS_CLOSED!!	April 16 2019 1:51
12	HALF	BOX_IS_CLOSED!!	April 16 2019 1:51

The caretaker and the doctor can also check the status of the medicine supply in real-time using the cloud web page. This helps the caretaker to get an idea of when the box may run out of medicines. The doctor can also keep a track of the stock available before prescribing any new medicaions.

CONCLUSION

This paper explains about a Smart Pill Box which can be used to improve the adherence of a patient to medications on a daily basis. It aims at elderly patients and patients

who have problems with remembering medicine dosages and time of taking medicines. All these problems are addressed using an array of sensors, IoT and cloud service. The Smart Pill contains an IR Module and an Ultrasonic sensor for gathering data. These sensors are attached to the box to sense if the box is open or closed. This gathered data is sent to the cloud server using ESP8266 Wi-Fi Module as a mode of connectivity. All the gathered data is thus stored in a cloud server which can be accessed by the caretaker and the doctor to monitor the medications of the patient in real-time. The SIM800C GSM Module is used to send SMS alerts to the patient and the caretaker.

It sends a reminder to the mobile number of the patient to take his medications 15 minutes prior to the time when the medicine is supposed to be taken. Then, an SMS alert is also sent to the caretaker regarding if the patient really took the medication. When the box is opened, an alert is sent to the caretaker. The other functionality of the GSM module is to send an alert to the patient when the supply of medications in low in the Smart Pill Box. When the ultrasonic sensor senses a critically low level of medication, it triggers I/O and an SMS alert is sent to the patient, so that he can restock his supply of medicines. This allows the patient to purchase the medicines before they actually run out. The Smart Pill Box dramatically reduces the occurrence of a patient missing his medications. This in-turn improves the efficiency of the effect of medicine on the disese.

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